

Author Response to Reviewer u34v

All changes marked in blue; line refs refer to revised `main.tex`.

R3.1: Too fragmented due to titled paragraphs. **Response:** Removed all bold/italic titled paragraphs and converted to flowing prose with topic sentences throughout:

- Related Work: 6 bold titles replaced with transitions [L170, 173, 175, 179, 182]
- Discussion: taxonomy [L598], property categories [L600], hierarchy [L602], simplicity [L604], prior work [L609], practical implications [L611]
- Limitations [L620], Reproducibility [L622], Appendix [L1037, 1039]

R3.2: Research questions as actual questions feels informal. **Response:** Rephrased RQ1–3 as declarative objectives [L147--149]. E.g., RQ1: “We characterize which molecular property types benefit from 3D conformer geometry and provide a mechanistic basis for the observed selectivity.”

R3.3: Performance hierarchy confused. **Response:** Substantially rewritten at [L602] with mechanistic explanation of four tiers in terms of preserved 3D relational structure: “DKO discards it into summary statistics, engineered descriptors preserve selected physical invariants, and GNNs retain the full atomic graph.” The progression from tier 4 to tier 1 is now explicit rather than presented as an unexplained ranking. The key result—FP+3D descriptors matching SchNet on ESOL (1.000 vs. 1.004)—highlighted as evidence that engineered features can approach end-to-end learning. Abstract [L82], Contribution 3 [L157], Conclusion [L618] updated.

R3.4: Text too direct; expand into technical discourse. **Response:** Expanded terse statements into technical language at multiple locations:

- Abstract: now specifies “continuous-filter and equivariant message passing” and “pre-computed feature bottleneck” [L82]
- Bias-variance: explicit parameter-count calculation (~ 180 vs. ~ 7 samples/param), connection to model selection theory [L605]
- GNN justification: technical comparison of continuous-filter vs. equivariant paradigms, why two architectures suffice [L453]
- Truncation: curse of dimensionality vs. information loss argument with empirical evidence [L508]
- Practical framework: specific thresholds (>500 molecules, solvation properties) and four-step decision rules [L611]
- Conclusion: precise statistical claims ($p < 10^{-4}$) and explicit practitioner recommendations [L618--620]